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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

AI-Assisted Smart Waste Collection and Recycling Management System

DSA0104-OBJECT ORIENTED PROGRAMMING WITH C++

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ABSTRACT

The **AI-Assisted Smart Waste Collection and Recycling Management System** is designed to improve waste management using smart technology and C++. The system monitors waste bins and identifies their waste levels. It helps classify waste into recyclable and non-recyclable categories. The system also plans waste collection and finds suitable routes to reduce time and fuel usage. The collected waste data is analyzed to support better recycling and waste management decisions. Overall, the system helps reduce pollution, improve recycling, and maintain a cleaner and healthier environment.

LITERATURE REVIEW

Year	Author(s)	Title	Method / Technology	Advantages	Limitations
2025	P. Latha et al.,	Smart Waste Management System	C++ + Sensors	Simple waste monitoring	Requires sensors
2024	M. Abdallah and M. Hosny	Automatic Waste Classification	C++ + Image Processing	Helps identify waste types	Classification may not be highly accurate
2024	M. S. B. Islam et al.	Smart Waste Collection	C++ + Queue	Organizes waste collection	Basic approach
2023	T. A. Tiwari et al.,	Waste Collection Route Optimization	C++ + Graph Algorithm	Finds shorter collection routes	Depends on input data
2023	H. A. Khan et al.,	Recycling Management System	C++ + Data Structures	Easy waste record management	Mainly software-based



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PROBLEM STATEMENT



- ❖ Waste bins often become overloaded because they are not monitored regularly.
- ❖ Unnecessary collection trips waste time, fuel, and money.
- ❖ Different types of waste are mixed together, making recycling difficult.
- ❖ Manual waste management is slow and less efficient.
- ❖ Improper waste disposal causes pollution and creates an unhealthy environment.



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PROPOSED OBJECTIVES

- Develop a **C++-based smart waste collection system**.
- Use **Rule-Based AI** to classify different types of waste.
- Identify **recyclable, organic, and non-recyclable waste**.
- Assign **collection priority** based on waste information.
- Manage requests through **add, view, search, update, and delete** operations.
- Store records using **file handling** and generate recycling reports.

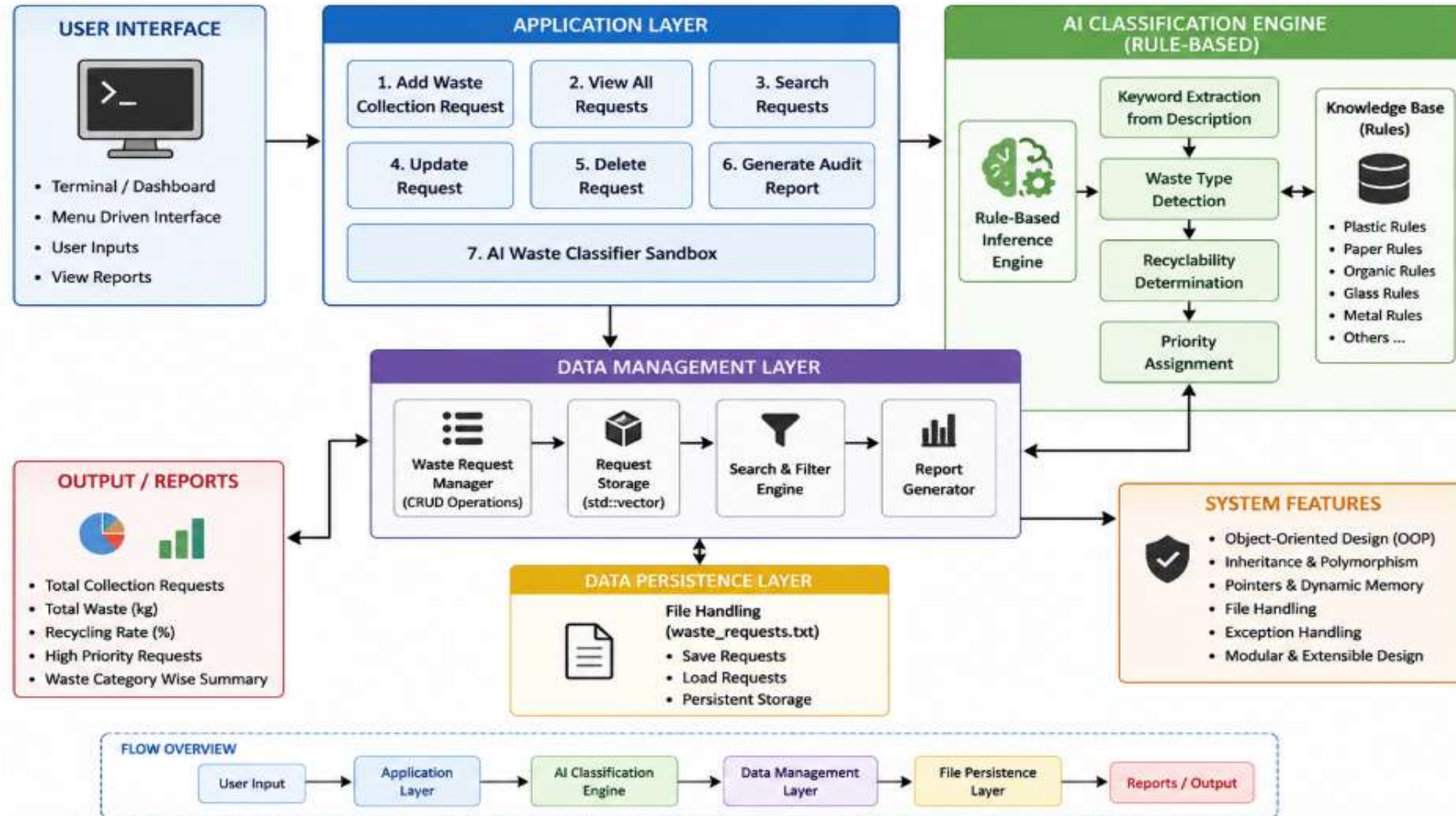


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ARCHITECTURE DIAGRAM



AI-Assisted Smart Waste Collection and Recycling Management System Architecture Diagram



MODULE 1 – SMART ECO BIN MONITORING

A) OBJECTIVE

- Monitor the waste level in each bin.
- Identify whether the bin is empty, half-full, or full.
- Store bin details and generate alerts for full bins.

B) METHODOLOGY

- Collect bin details such as bin ID, location, and waste level.
- Classify the bin as **Empty, Half-Full, or Full**.
- Generate an alert when the bin becomes full.

C) TECHNOLOGY

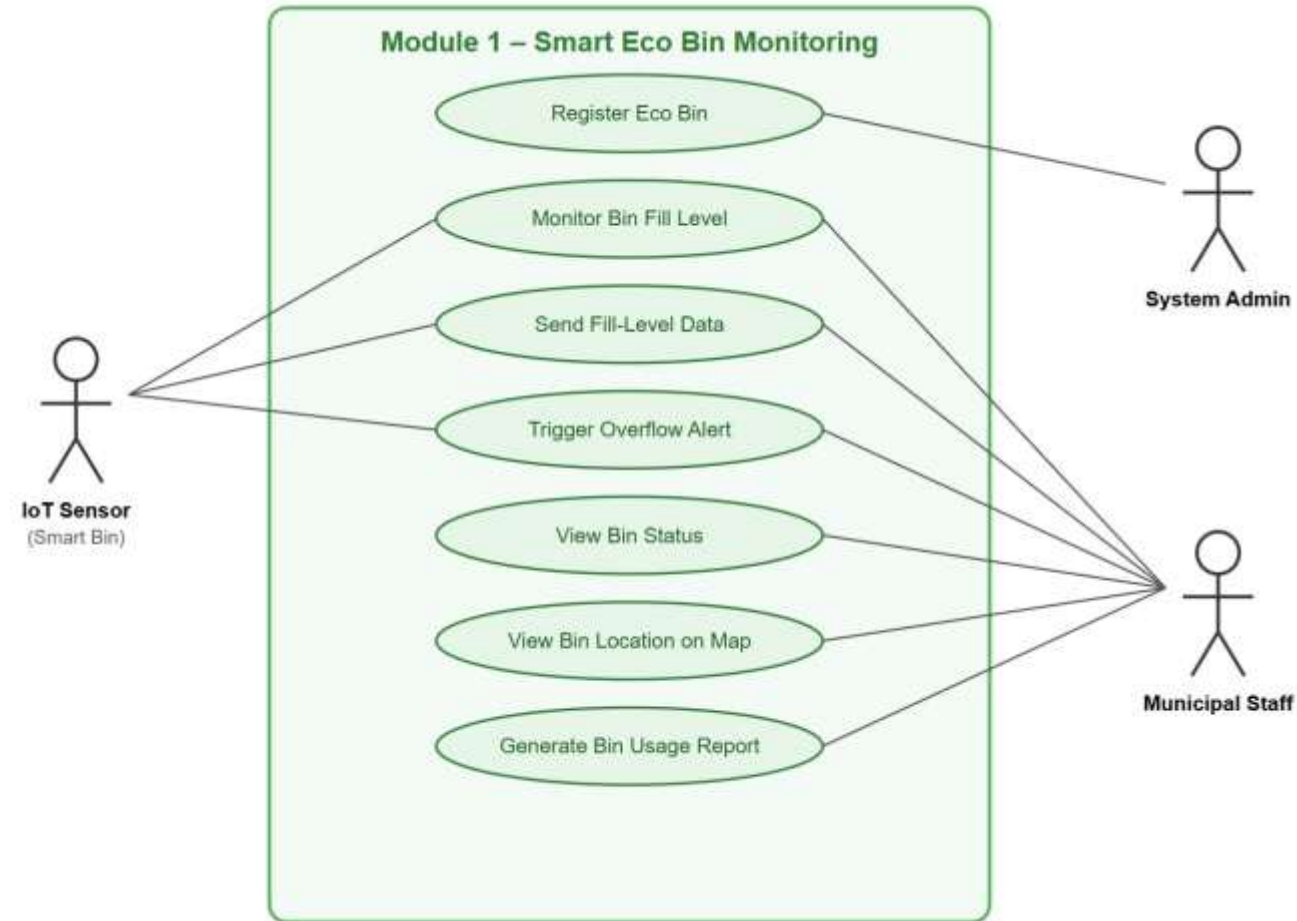
- C++
- Structures
- Arrays
- Conditional Statements



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USECASE DIAGRAM



MODULE 2 – AI WASTE INTELLIGENCE ENGINE

A) OBJECTIVE

- Identify different types of waste.
- Classify waste as recyclable or non-recyclable.
- Support better waste segregation and recycling.

B) METHODOLOGY

- Take waste details as input.
- Identify the category of the waste.
- Separate waste into recyclable and non-recyclable types.
- Store the classification result.

C) TECHNOLOGY

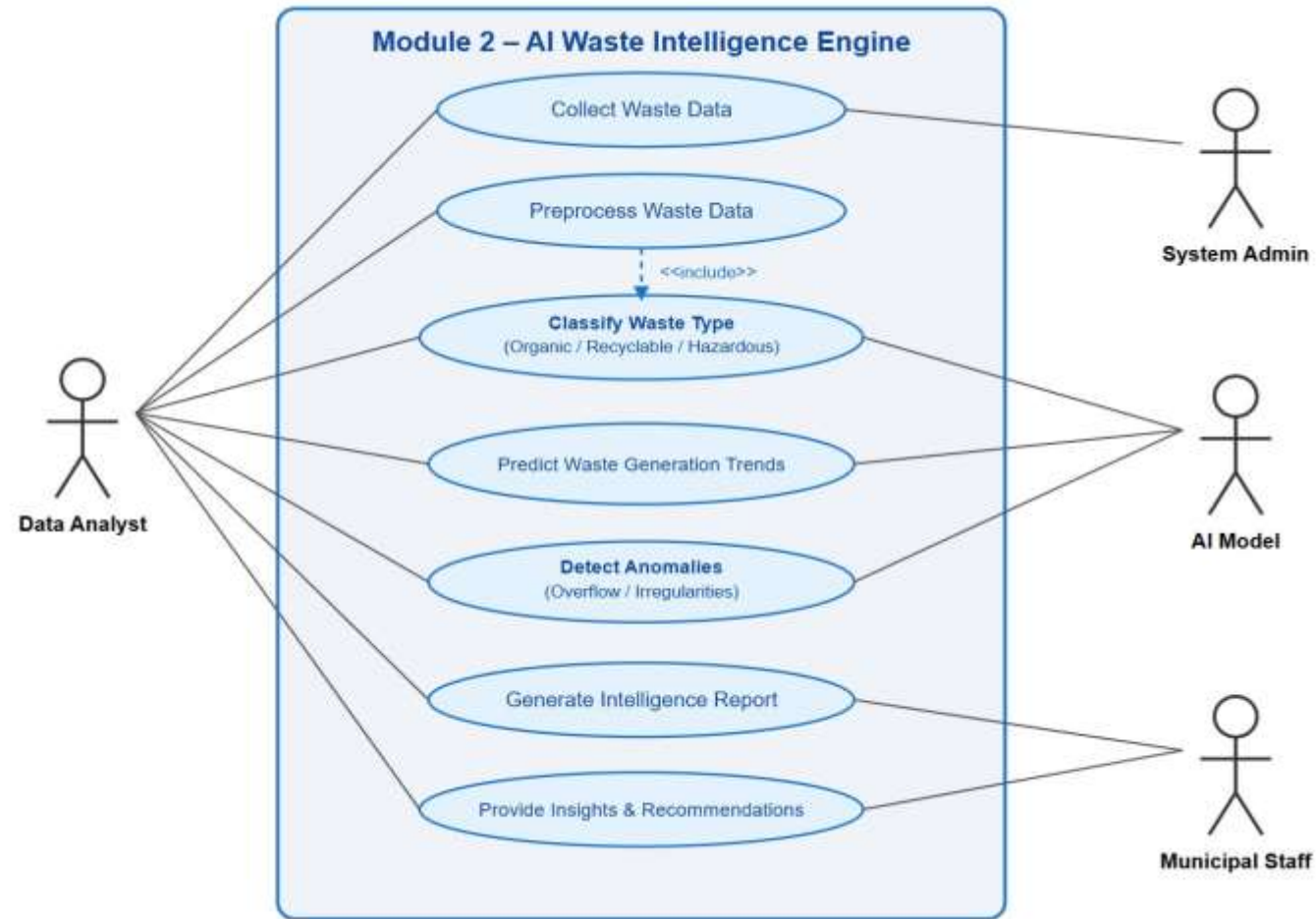
- C++
- Decision Tree Algorithm
- String Processing



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USECASE DIAGRAM



MODULE 3 – SMART COLLECTION & GREEN ANALYTICS

A) OBJECTIVE

- Plan waste collection based on bin status.
- Find efficient collection routes.
- Analyze waste data to reduce fuel usage and pollution.

B) METHODOLOGY

- Select bins that require collection.
- Assign priority based on waste level.
- Find the shortest collection route.
- Calculate collection distance and efficiency.

C) TECHNOLOGY

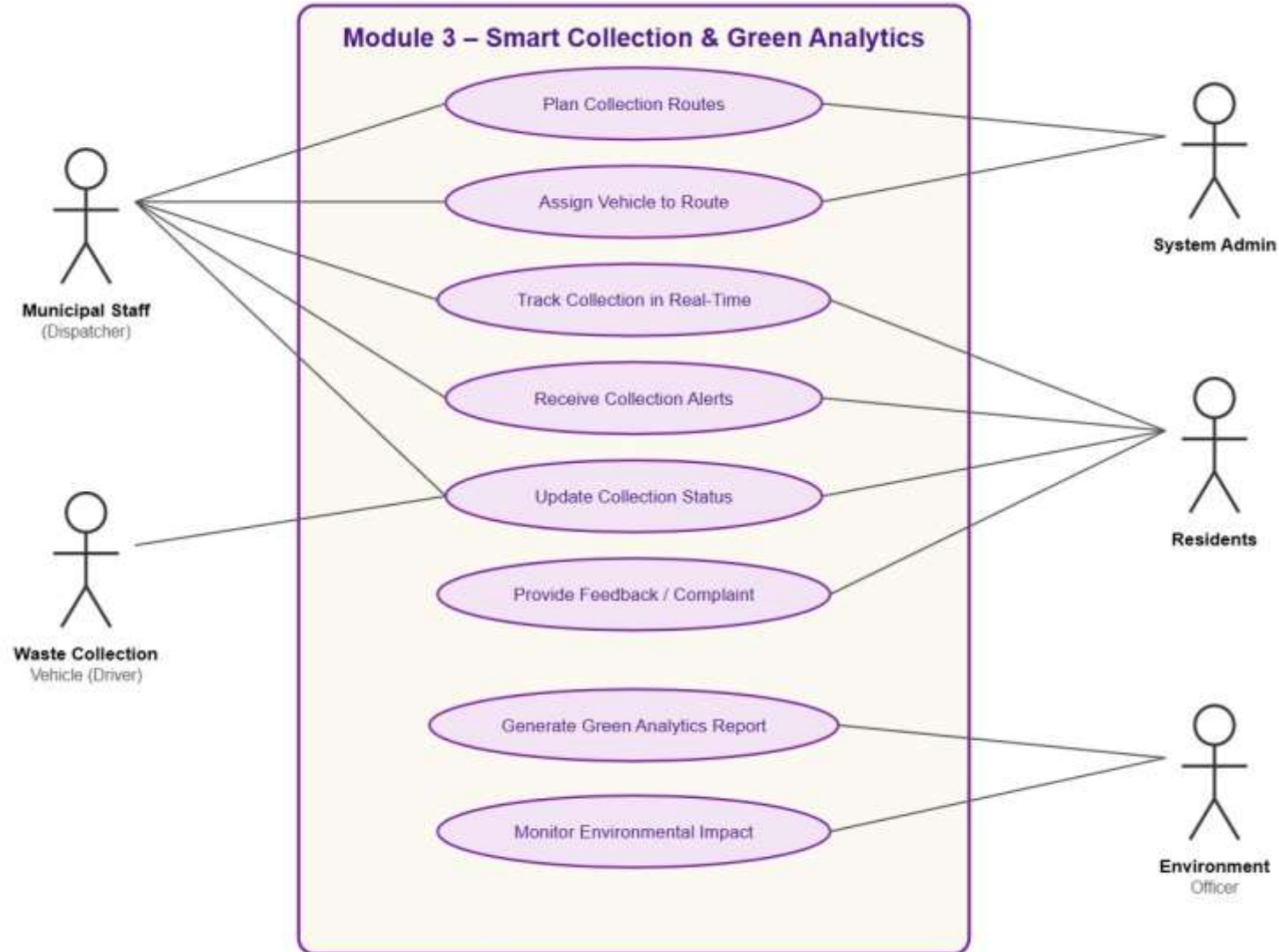
- **C++**
- **Queue** for collection priority
- **Graph Algorithm** for route planning
- **Dijkstra's Algorithm** for shortest path



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USECASE DIAGRAM





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IMPLEMENTATION

AI-Assisted Smart Waste Collection and Recycling Management System

Remix Device

Requests Monitor & C++ Terminal v1.0 C++17

Search location...

+ New Request Download .cpp

SmartWaste AI
C++ Management System

Dashboard & Terminal

C++ Source Code

Classification Logic

Sample I/O Tests

Compilation Guide

TOTAL REQUESTS:
7
std::vector memory instances

RECYCLING RATE:
98.8%
Landfill diversion efficiency

TOTAL MASS:
164.7 kg
Organic: 10.2 kg

PRIORITY HIGH:
2
Urgent collection tickets

CONSOLE SHORTCUTS 1-Click Dispatch

- 1 Add Request
AI Auto-Classify
- 2 View All
Virtual table
- 3 Search
ID / Location
- 4 Audit Report
CO2 & Diversion
- 5 Standalone AI Classifier Sandbox
Test keyword rules without saving to vector

VECTOR HEAP MEMORY 7 objects allocated

Polymorphic base pointers allocated on heap via new
DerivedClass() and registered into
std::vector<WasteRequest*>

```
// Memory Footprint:
sizeof(WasteRequest*) = 8 bytes
Persistence: waste_requests.txt [OK]
```

[Zero-API Rule-Based AI Engine & Object-Oriented Framework]

[SYSTEM] Initialized WasteManager controller with local persistence.
[SYSTEM] 5 initial polymorphic waste request records loaded into std::vector<WasteRequest*>.

MAIN NAVIGATION MENU

1. Add Waste Collection Request (with Rule-Based AI Auto-Classification)
2. View All Waste Collection Requests (Polymorphic Details)
3. Search Requests (by ID, Category, or Keyword)
4. Update Waste Collection Request
5. Delete Waste Collection Request
6. Generate Smart Recycling & Environmental Audit Report
7. Standalone AI Waste Classifier Sandbox (Interactive Test)
8. Save Data & Exit System

Enter selection (1-8):

user@waste-cpp:~\$ Type 1-8 and press Enter (e.g. 1 to Add, 8 for Report)...

Send

RESULTS

- **Waste Classification:** The system successfully classifies waste into recyclable, organic, and non-recyclable categories.
- **Priority Detection:** The system identifies bins as High, Medium, or Low priority based on their waste level.
- **Collection Recommendation:** The system recommends which bins should be collected first and generates a simple waste management report.



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WASTE-BASED CLASSIFIER

```
if (waste == 'plastic')
    add (code == 1)
    then 'Plastic' (PETH)
```

Standard C++11 · OOP Dynamic

Active Collection Requests 7 of 7

All Recyclable Landfill Urgent Filter table...

ID	TYPE	STATUS	QUANTITY	LOCATION & ITEM DESCRIPTION	PRIORITY
#181	Plastic	RECYCLABLE	24.5 kg	Sector 4 Community Center PET Mineral Water Bottles & Clean Caps	MEDIUM
#182	Paper	RECYCLABLE	48.0 kg	Warehouse Dock B Compressed Shipping Cartons and Bulk Paper	HIGH
#183	Organic	RECYCLABLE	18.2 kg	Central Cafeteria Kitchen Vegetable Peels, Coffee Grounds & Food Scraps	HIGH
#184	Glass	RECYCLABLE	35.0 kg	Downtown Streets Alley Soda/Alcohol Beverage Glass Bottles	MEDIUM
#185	Metal	RECYCLABLE	12.0 kg	North Campus Hall Aluminum Tidy Cans and Clean Foil Tray	LOW
#186	Plastic	LANDFILL	2.0 kg	2	MEDIUM
#187	Plastic	RECYCLABLE	25.0 kg	Chennai Waste Type: Plastic Quantity: 25 Locations: Chennai	LOW

Total Glass: 35.0kg · Total Plastic: 51.5kg · Total Metal: 12.0kg · Total Paper: 48.0kg

System Status: Active

System Algorithms & Rule-Based AI Engine

Zero dependency heuristic inference, object-oriented dynamic dispatch and metric correlations

Live Heuristic AI Classifier Sandbox

Enter Waste Item Description (NLP Pattern Input):

used greasy pizza box

Weight (kg):

12

Classified Category: Paper

Recyclability: **NON-RECYCLABLE**

Assigned Priority: **[LOW]**

Rule Triggered: AS Rule #302: Paper contaminated with grease/moisture impairs pulp fibers. Non-recyclable.

Handling Advice: Tear off clean lid for recycling! Discard greasy sections with general waste.

STEP-BY-STEP ALGORITHMIC LOGIC

1 System Initialization & Data Rehydration

On startup, instantiate the WasteManager controller. Check if 'waste_requests.txt' exists; if found, parse each line via pipe delimiter tokens, reconstruct polymorphic objects ('new PlasticWasteRequest', etc.) and populate the vector 'requests'. If not found, seed initial realistic records.

PSEUDOCODE LOGIC

```
FUNCTION InitializeSystem()
    OPEN "waste_requests.txt"
    IF file exists THEN
        FOR EACH line IN file:
            tokens = split(line, "|")
            reqPtr = factory.create(tokens[0..5])
            requests.push_back(reqPtr)
        ELSE
            seedInitialRecords()
    END IF
END FUNCTION
```

INTERACTIVE CONCEPT INSPECTOR (CLICK TO HIGHLIGHT CODE BLOCK):

Classes & Objects

OOP Fundamental

Constructors & Destructors

Memory Lifecycle

Inheritance Hierarchy

Code Reusability

Polymorphism & Virtual Functions

Dynamic Dispatch

Vectors & Dynamic Pointers

Heap Management

File Handling & Persistence

Stream I/O

Exception Handling

Robustness

Rule-Based AI Classifier

Heuristic Inference

CONCLUSION

- The system automates waste classification, determines recyclability and collection priority, manages collection requests, maintains persistent records, and generates environmental reports. The implementation also demonstrates important C++ concepts such as object-oriented programming, inheritance, polymorphism, pointers, vectors, and file handling.



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FURTHER SCOPE

- **IoT Integration** – Connect smart bins with sensors to monitor waste levels automatically.
- **Mobile Application** – Develop a mobile app to show bin status and collection updates.
- **Advanced AI** – Use machine learning to identify different waste types more accurately.
- **GPS Tracking** – Use GPS to track collection vehicles and provide real-time routes.
- **Smart Recycling** – Improve automatic waste separation and increase recycling efficiency.





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REFERENCES



1. P. Latha et al., “Smart and efficient waste management through wireless IoT-enabled deep learning,” *Scientific Reports*, 2026.
2. M. Abdallah and M. Hosny, “Development and simulation of an innovative autonomous knowledge-based smart waste collection system,” *Scientific Reports*, 2026.
3. M. S. B. Islam et al., “ECCDN-Net: A deep learning-based technique for efficient organic and recyclable waste classification,” *Waste Management*, 2025.
4. T. A. Tiwari et al., “Smart waste management system using AR and IoT with AI,” in *Proc. Int. Conf. Artificial Intelligence and Data Engineering*, 2025.
5. H. A. Khan et al., “Enhancing trash classification in smart cities using federated deep learning,” *Scientific Reports*, 2024.



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THANK YOU